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Lindsay M. Smith

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Research Interests

My research focuses on the **science of AI** and **AI for science**. I apply techniques and ideas from physics to artificial neural networks to understand how they learn, and I develop methods for AI to learn to conduct science effectively. My current research in machine learning and AI includes projects in **interpretability**, **AI for scientific discovery**, **in-context learning**, **LLM multi-agent interactions**, and **AI safety**.

Education

M.A., Ph.D. — Princeton University — Physics 2022 - 2027

NSF GRFP

Advisors: William Bialek and David Schwab.

B.A. — University of Pennsylvania — Physics (Honors) 2018 - 2022

Advisor: Dani Bassett.

Cum Laude; Minors in Mathematics and French & Francophone Studies.

Experience

Junior Research Scientist — Polymathic AI Mar 2026 - Aug 2026

Internship Project: AI for Scientific Discovery (paper under review at NeurIPS 2026).

Mentee — MARS (Mentorship for Alignment Research Students) Jul 2025 - Feb 2026

MARS 3.0 participant at the Cambridge AI Safety Hub. Co-first author on project investigating AI control and steering via Chain-of-Thought (CoT) injections (paper at workshop at ICLR 2026).

Graduate Researcher — Princeton University Aug 2022 - Present

Awards & Honors

Citadel GQS Physics PhD Fellowship Finalist (3 finalists nationally) 2026

National Science Foundation (NSF) ACCESS Computing Award 2025

(200,000 credits, ~ \$5,000)

Applied for and am Principal Investigator for this award.

NSF AI Institutes Virtual Organization (AIVO) Travel Grant (\$2,000) 2025

American Physical Society (APS) GSNP Student Speaker Award Finalist (\$500) 2025

NSF Graduate Research Fellowship Program (GRFP) (\$159,000 + \$12,000 bonus from Princeton) 2022 - 2027

Charlotte and Morris Tanenbaum *52 Graduate Fellowship in Physical or Life Sciences (\$40,000) 2022 - 2023

Joseph Henry Merit Award (\$5,000) 2022

University Scholars Program (\$7,000) 2020 - 2022

Applied for and awarded summer research funding in 2020 and 2021.

National French Honor Society – Pi Delta Phi 2020 - Present

Sister Loretta Thome Scholarship (\$10,000) 2018 - 2023

Publications [Google Scholar] * indicates equal contribution.

9. Matt L. Wiemann*, **Lindsay M. Smith***, Peter Melchior, Siddharth Mishra-Sharma, Andrew Gordon Wilson, Pavel Izmailov, Carolina Cuesta-Lázaro (2026). DISCOVERPHYSICS: Benchmarking LLMs for Out-of-the-Box Scientific Thinking. *Under review*, <https://arxiv.org/abs/2605.26087>.

8. **Lindsay M. Smith***, Ananya Malik*, Edward James Young, Puria Radmard, Cameron Tice, and Hannes Whittingham (2026). Chain-of-Thought Injection as an Inference-Time Safety Intervention. *ICLR 2026 Workshop on Logical Reasoning of LLMs*, <https://openreview.net/forum?id=v0XkgeD6U>.
7. Alireza Karami, Veronica Chelu, Po-Chen Kuo, **Lindsay M. Smith**, Mia Whitefield, Jonathan D Cohen (2026). A Computational Basis of Natural Intelligence. *Proceedings of the Analytical Connectionism Schools 2023–2024*, *PMLR* 320:151-176
6. Colin Scheibner*, **Lindsay M. Smith***, and William Bialek (2025). Large language models and the entropy of English. *Under review*, <https://arxiv.org/abs/2512.24969>.
5. Jeff Shen* & **Lindsay M. Smith*** (2025). ALICE: An Interpretable Neural Architecture for Generalization in Substitution Ciphers. *Under review*, <https://arxiv.org/abs/2509.07282>.
4. Chase Goddard, **Lindsay M. Smith**, Vudtiwat Ngampruetikorn*, David J. Schwab* (2025). When can in-context learning generalize out of task distribution? *ICML 2025*, <https://arxiv.org/abs/2506.05574>.
3. **Lindsay M. Smith**, Chase Goddard, Vudtiwat Ngampruetikorn*, David J. Schwab* (2024). Model Recycling: Model component reuse to promote in-context learning. *NeurIPS 2024 Workshop on Scientific Methods for Understanding Deep Learning*, <https://openreview.net/forum?id=vWSu8nEURM>.
2. Chase Goddard, **Lindsay M. Smith**, Vudtiwat Ngampruetikorn*, David J. Schwab* (2024). Specialization-generalization transition in exemplar-based in-context learning. *NeurIPS 2024 Workshop on Scientific Methods for Understanding Deep Learning*, <https://openreview.net/forum?id=D1ui5QwHqF>.
1. **Lindsay M. Smith**, Jason Z. Kim, Zhixin Lu, and Dani S. Bassett (2022). Learning continuous chaotic attractors with a reservoir computer, *Chaos* 32, 011101, <https://doi.org/10.1063/5.0075572>. *Selected as an Editor’s Pick and publicized with a Scilight summary: <https://doi.org/10.1063/10.0009079>*.

Talks

Kempner Institute Workshop: Learning Dynamics in Natural & Artificial Intelligence , Cambridge, MA	Sep 2026
Talk: “Learning long-ranged structure in English: Code length and entropy dynamics during LLM training.”	
APS Global Summit , Denver, CO	Mar 2026
Talk: “Universality of LLM mechanisms across scale and diversity.”	
Princeton University Machine Learning Journal Club , Princeton, NJ	Sep 2025
Invited talk: “ALICE: An Interpretable Neural Architecture for Generalization in Substitution Ciphers.”	
APS Global Summit , Anaheim, CA	Mar 2025
<i>(GSNP Student Speaker Award Finalist)</i>	
Talk: “Multi-Agent Debate: Analyzing Consensus in Networks of LLM Agents.”	
APS March Meeting , Chicago, IL	Mar 2022
Talk: “Learning Continuous Chaotic Attractors with a Reservoir Computer.”	

Posters

Princeton Center for Theoretical Science Workshop: The Physics of John Hopfield , Princeton, NJ	Sep 2025
“ALICE: An Interpretable Neural Architecture for Generalization in Substitution Ciphers.”	
Meta 2025 PhD Forum , Menlo Park, CA	Oct 2025
“ALICE: An Interpretable Neural Architecture for Generalization in Substitution Ciphers.”	

NeurIPS 2024 Workshop on Scientific Methods for Understanding Deep Learning , Vancouver, Canada	Dec 2024
“Model Recycling: Model component reuse to promote in-context learning.”	
Conference for Undergraduate Women in Physics (CUWiP) , Virtual	2022
“Learning Continuous Chaotic Attractors with a Reservoir Computer.”	
Penn Research Expo , Philadelphia, PA	2020, 2021, 2022
“Development of control in brain networks over temporal and spatial scales using graph models.”	
“Learning Continuous Chaotic Attractors with a Reservoir Computer.”	

Teaching and Mentoring

Teaching Assistant , Princeton University	Fall 2025
PHY 103: General Physics I	
Mentor , Princeton Towards an Inclusive Physics cOmmunity in graduaTe Students (TIPOTS)	2024 - Present
Mentored younger physics graduate students each semester.	
Mentor , Princeton Physics Mentorship	2022 - 2024
Mentored one to two undergraduate physics students each semester. Met at least once a semester to give career and academic advice.	
Tutor , CIS 110: Introduction to Computer Programming	2021 - 2022
Tutored two to three students weekly.	
Mentor , Side By Side Agency	2021
Advised a student on an independent astrophysics research project and poster.	

Professional Service and Outreach

Reviewer: Workshop on Scientific Methods for Understanding Deep Learning (Sci4DL) at NeurIPS 2024 and ICLR 2026.

Princeton Women in Physics, **Executive Board** (2023 - Present)

Princeton Physics EDI, **Events Committee** (2022 - 2024)

Technical Skills

Python, PyTorch, JAX, L^AT_EX, bash, Git, Jupyter, MATLAB, Java, C++, ROOT, Mathematica, Linux